Ian Overturf

Software Engineer | Gameplay & Engine Systems
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SKILLS

Languages: C++, C#, C, Java, Python, Lua, HLSL, x86 Assembly
Game Dev: Unreal Engine 4 & 5, Unity, Godot, DirectX
Platforms: Windows, PS5 / PS5 Pro, PS4, Xbox X/S, Xbox One, Android
Tools: Visual Studio, VS Code, Rider, Windsurf, Perforce, Git, Jenkins, Jira, Helix ALM

WORK EXPERIENCE

Software Engineer - 2K Games (Visual Concepts) *Jan. 2023 - Apr. 2026*
Austin, TX

R&D: Unreal Engine 5 - Gameplay, Crowd, Rendering Systems (Unannounced Title / ~2 Years)

- Architected a large-scale crowd system in UE5 which supported up to 5,000 dynamically animated crowd members
- Built a VAT (Vertex Animation Texture) bake pipeline via AnimToTexture with full per-character wardrobe variation
- Implemented behavior and animation logic for autonomous AI agents within the crowd
- Reduced VAT texture data from ~75GB to ~2GB on disk through bone compression and shared-skeleton VAT deduplication, with runtime memory savings of ~5-10GBs
- Created a C++ plugin framework to drive gameplay behavior and character rendering via an external third-party DLL

R&D: NBA AI Engine Integration - (C++ / ~3 Years)

- Contributed to core development of a C++ DLL bridge to run NBA 2K in external engines, later taking over as the sole maintainer after the initial architecture phase
- Owned daily production merges and conflict resolution across gameplay, AI, and library changes
- Drove a large-scale architectural refactor of a dedicated NBA 2K branch to decouple it from production dependencies, with the goal of merging this system back to main
- Exposed critical data (messaging, hooks, state) to third-party users of this system to support their development
- Maintained PS5 and PS5 Pro build support; diagnosed crashes, memory leaks (including static global persistence across restarts), and thread race conditions
- Developed proof of concept builds to illustrate multiple AI instances for server/gameplay development

NBA 2K25 - Gameplay, Console

- Resolved long-standing legacy certification blockers (Xbox XR / Sony TRC) that had required recurring annual waivers, eliminating the need for ongoing prior approval cycles
- Integrated Live Service content drops (battle pass, clothing, art) and validated delivery in Sandbox/Live environments
- Collaborated with QA and production to identify and resolve critical gameplay and certification issues under a tight ship deadline

NBA 2K24 - Gameplay

- Implemented challenge verification logic for a licensed IP challenge mode, tracking and validating in-game player performance against specific scoring and accuracy thresholds
- Backported complex gameplay systems from upstream to the older legacy codebase, adapting logic and resolving compatibility issues to maintain feature parity across generations
- Designed and prototyped a first-time user experience minigame to onboard new and returning players to updated shooting mechanics, balancing accessibility with engagement

LEGO 2K Drive - Gameplay & UI

- Diagnosed and resolved issues around minigames/racing related to score tracking, camera behavior, and race state
- Took primary ownership of the in-game storefront for gameplay/microtransaction flows, including live service features such as entitlements, deep linking, and redemption codes
- Eliminated a major performance bottleneck by replacing full catalog rebuilds with a single initialization pass and targeted per-item updates on purchase
- Developed new UI/UX components for shop navigation, purchasing, and storefronts

EDUCATION

DePaul University *Chicago, Illinois*
B.S. in Computer Science (Game Systems) Graduated March 2022
Minor in Game Design, Summa Cum Laude - 3.96 GPA